

Lab No: 01

Experiment Title: Arithmetic Operations in Assembly Language Programming Using EMU8086

Objective

The objective of this experiment was to understand and implement basic arithmetic operations in assembly language using EMU8086 software. Specifically, the experiment aimed to apply multiplication, division, and addition operations to convert temperature values from Celsius to Fahrenheit using the provided mathematical expression.

Theory

Assembly language is a low-level programming language that provides direct communication with the processor through machine-level instructions. Unlike high-level languages, assembly language enables programmers to perform operations at the hardware level by directly manipulating registers and memory locations.

In this experiment, arithmetic operations were implemented to convert temperature measurements from Celsius (°C) to Fahrenheit (°F). The mathematical relationship between these temperature scales was expressed using the formula:

$$F = (C \times 9/5) + 32 + 1$$

Where C represented the input temperature in Celsius and F represented the output temperature in Fahrenheit. The calculation was performed sequentially using three main arithmetic operations: multiplication, division, and addition. These operations were executed in the order specified to ensure accuracy in the temperature conversion.

Assembly Language Code

The screenshot displays the EMU8086 software interface. On the left, the assembly code is shown, and on the right, the extended value viewer displays the results of the operations.

Assembly Code:

```
01  
02 ORG 100H  
03  
04 .DATA  
05 C DW 25  
06 F DW ?  
07  
08 .CODE  
09 MAIN PROC  
10     MOV AX, @DATA  
11     MOV DS, AX  
12     ; F = (C * 9/5) + 32 + 1  
13  
14     MOV AX, C  
15     MOV BX, 9  
16     MUL BX  
17  
18     MOV BX, 5  
19     DIV BX  
20  
21     ADD AX, 32  
22     INC AX  
23  
24     MOV F, AX  
25  
26 MAIN ENDP  
27 END MAIN  
28
```

Extended Value Viewer:

H		L	
hex:	00	4E	
bin:	00000000	01001110	
oct:	000	116	
decimal 8 bit			
unsigned:	0	78	
signed:	0	78	
ascii:		N	
decimal 16 bit			
unsigned:	78		
signed:	78		

Calculation

Given:

$$C = 25^{\circ}\text{C}$$

Using the expression: $F = (C \times 9/5) + 32 + 1$

Substituting $C = 25$:

Step 1: Multiplication

$$25 \times 9 = 225$$

Step 2: Division

$$225 \div 5 = 45$$

Step 3: Addition

$$45 + 32 = 77$$

Step 4: Increment Operation

$$77 + 1 = 78$$

Output

Parameter	Value
Input Temperature	25°C
Output Temperature	78°F

Conclusion

The experiment was successfully completed using EMU8086 software. The assembly language program was executed to convert the input temperature value of 25°C to Fahrenheit. The arithmetic operations were performed sequentially in the correct order: multiplication of the Celsius value by 9, division by 5, addition of 32, and increment by 1. Each operation was successfully executed in the processor registers, and the final result of 78°F was correctly stored in the output variable. This experiment demonstrated the practical application of basic arithmetic operations in low-level assembly language programming and provided a clear understanding of how processors handle mathematical calculations at the hardware level.